

Assignment 3 (15% of total marks)

**Due date: Saturday, 28 February 2026 by 9:00 pm (21:00 hours)
Singapore time.**

Scope

The tasks of this assignment cover the **data structure and algorithm**. Focusing on greedy algorithm and algorithm design. The assignment covers the topics discussed in **Topic 8**, and **Topic 9**.

The assignment consists of three theory questions. Marks for each question are indicated next to the respective question. The total mark for Assignment 3 is **100 marks**.

Marks Distribution

- **Question 1:** 25%
- **Question 2:** 30%
- **Question 3:** 45%
- **Total Marks:** 100
- **Weightage:** 15% of the total subject mark

Assessment Criteria

Marks will be awarded for:

- **Correctness:** Solutions must meet the requirements of the question.
- **Comprehensiveness:** Solutions should demonstrate a complete understanding of the topic.
- **Appropriate Application:** Answers should reflect proper application of the materials covered in this subject.

Deliverables

Theory Questions (Questions 1 to 3)

- Type your answers using **MS Word** and save the completed document as a **PDF file**.
- Alternatively, you may write your answers by hand onto a piece of paper or using tablet. Ensure your handwriting is **neat**, then scan or save your handwritten solutions into a **PDF file**.

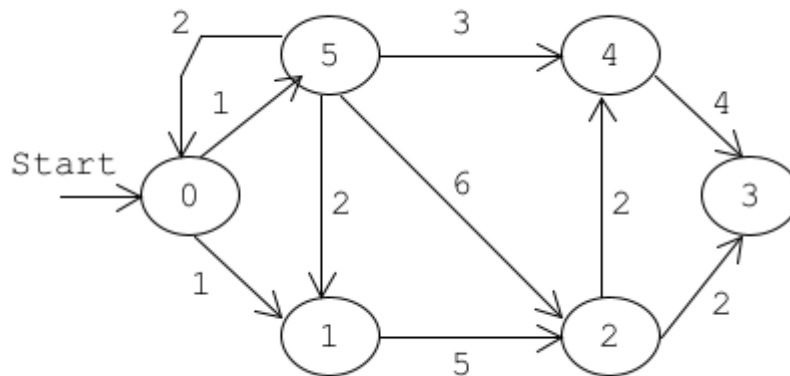
Submission Instructions

- If you have more than one PDF files, combine all the PDF files into a single **ZIP file** named **SolutionA3.zip**, and submit through Moodle.
- Do not include subdirectories in your submission.

Theory Questions (100 Marks)

Question 1 – Graph (25 Marks)

a. Consider a directed graph below:



Represent the graph as:

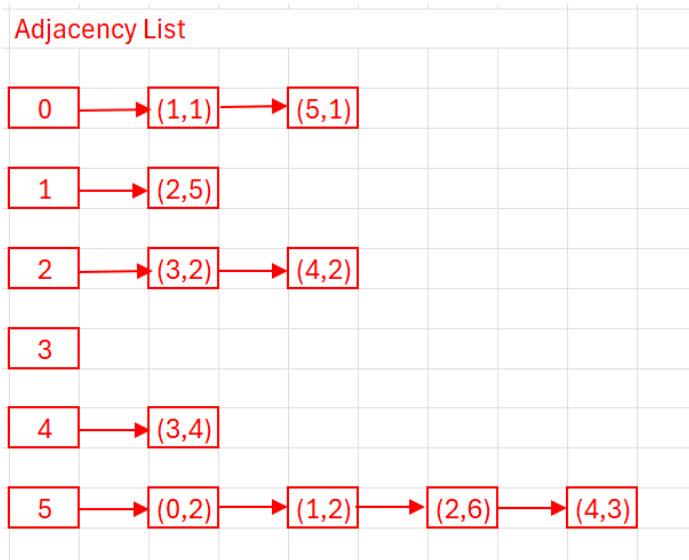
(i) Adjacency matrix

(5.0 Marks)

Adjacency Matrix							
		From					
To		0	1	2	3	4	5
	0	∞	∞	∞	∞	∞	2
	1	1	∞	∞	∞	∞	2
	2	∞	5	∞	∞	∞	6
	3	∞	∞	2	∞	4	3
	4	∞	∞	2	∞	∞	∞
	5	1	∞	∞	∞	∞	∞

(ii) Adjacency list

(5.0 Marks)



(iii) Incidence matrix

(5.0 Marks)

Incidence Matrix

	e_{0-1}	e_{0-5}	e_{5-0}	e_{5-1}	e_{1-2}	e_{5-2}	e_{5-4}	e_{2-4}	e_{4-3}	e_{2-3}
0	1	1	∞	∞	∞	∞	∞	∞	∞	∞
1	∞	∞	∞	∞	5	∞	∞	∞	∞	∞
2	∞	∞	∞	∞	∞	∞	∞	2	∞	2
3	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
4	∞	∞	∞	∞	∞	∞	∞	∞	7	∞
5	∞	∞	2	2	∞	6	3	∞	∞	∞

b. Using the same graph shown above:

- (i) Perform DFS (Depth First Search) starting from '0' and show the order of traversal. **(5.0 Marks)**

Sample solution includes but is not limited to the following:

DFS:

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Note: Assumption – When there are more than one paths from a node, scan/traverse the paths from right to left.

- (ii) Perform BFS (Breadth First Search) starting from '0' and show the order of traversal. **(5.0 Marks)**

Sample solution includes but is not limited to the following:

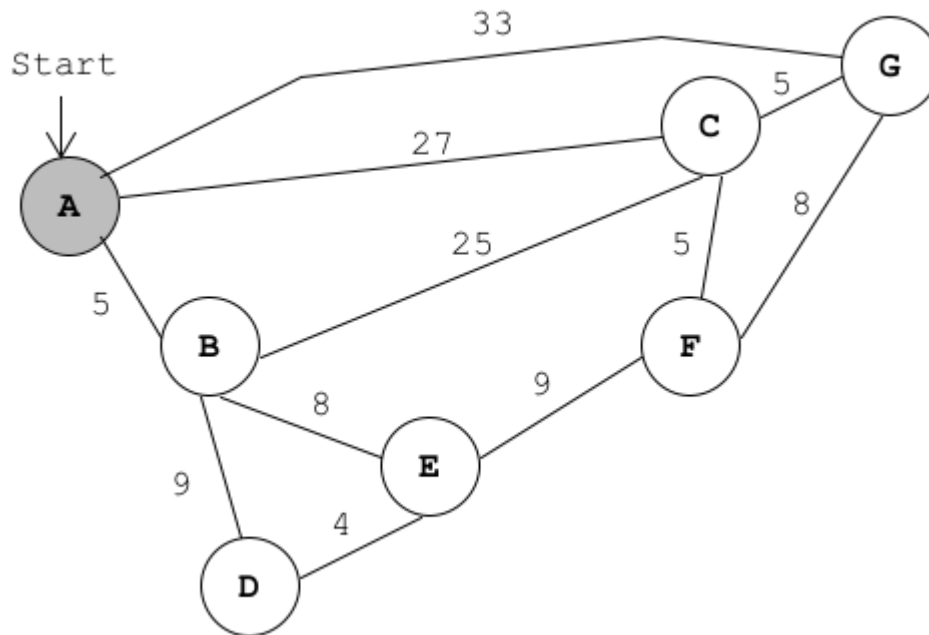
BFS:

Note: When there are more than one paths from a node, scan/traverse the paths from right to left.

$0 \rightarrow 5 \rightarrow 1 \rightarrow 4 \rightarrow 2 \rightarrow 3$

Question 2 – Queue and Stack (30 Marks)

Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from A to all network nodes.



- a) Show how the algorithm works by computing a table like the one discussed in class. **(25.0 marks)**

Sample Solution includes but is not limited to the following:

CSCI203-2026S1-A3-MarkingScheme
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From		↓		↓	↓		P	V	D	
To	0	1	2	3	4	5				s (0,0)(5,1)(1,2)(4,4)
0						2	0	0	2	
1	2					2	0	1	2	
2		5				6	5	2	7	
3			2		4		4	3	8	
4			2			3	5	4	4	
5	1						0	5	1	

From		↓	↓		↓	↓	P	V	D	
To	0	1	2	3	4	5				s (0,0)(5,1)(1,2)(4,4)(2,7)
0						2	0	0	2	
1	2					2	0	1	2	
2		5				6	5	2	7	
3			2		4		4	3	8	
4			2			3	5	4	4	
5	1						0	5	1	

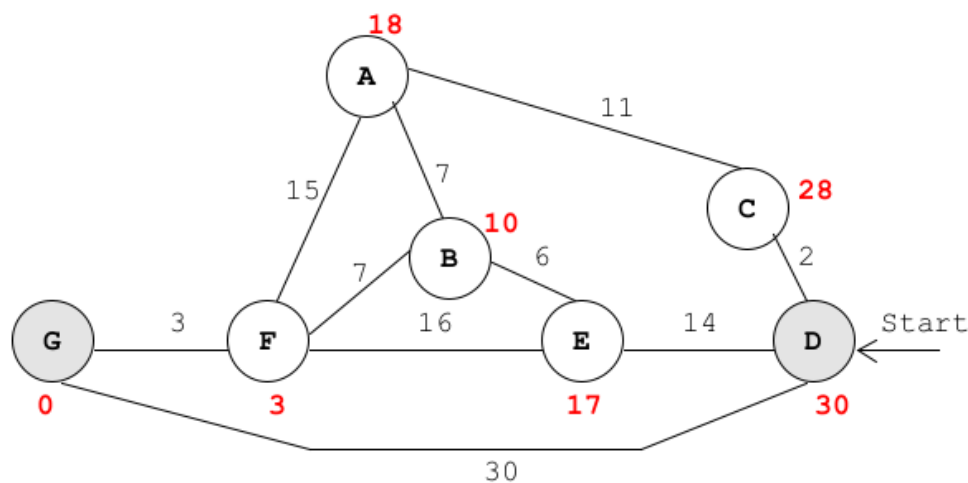
From		↓	↓		↓	↓	P	V	D	
To	0	1	2	3	4	5				s (0,0)(5,1)(1,2)(4,4)(2,7)(3,8)
0						2	0	0	2	
1	2					2	0	1	2	
2		5				6	5	2	7	
3			2		4		4	3	8	
4			2			3	5	4	4	
5	1						0	5	1	

b) Show all the paths from A to all other network nodes. **(5.0 marks)**

Shortest Path:			
0	1	:	0-1 (2)
0	2	:	0-5-2 (7)
0	3	:	0-5-4-3 (8)
0	4	:	0-5-4 (4)
0	5	:	0-5 (1)
Total Distance: (2 + 7 + 8 + 4 + 1 = 22)			

Question 3 – A* Heuristic Search (45 Marks)

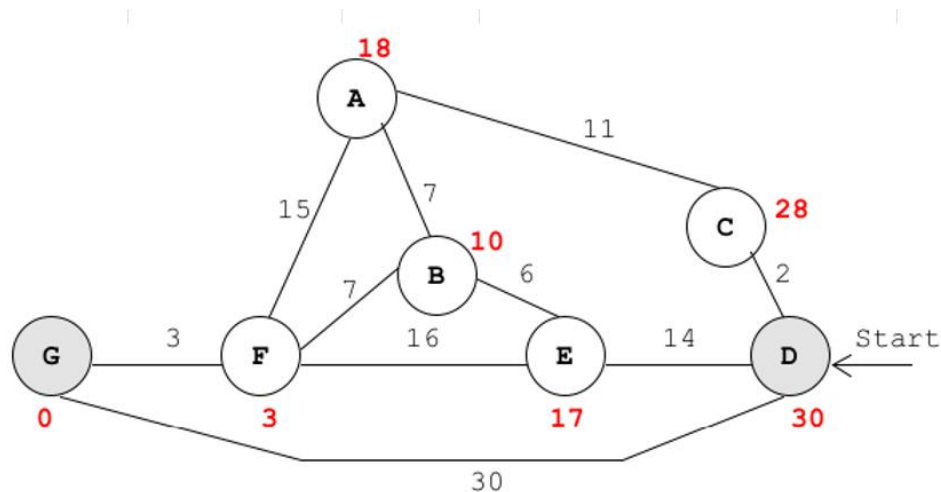
Consider the following search problem, represented as a graph. Each node is label by a capital letter, and the value of a heuristic function is shown in maroon. Each edge is labelled by the cost to traverse that edge. The start state is 'A' and the only goal state is 'G'. Perform the A* search to find the shortest path from node A to node G.



Number in maroon are the heuristics for the respective nodes.
 Number in black are the weight (estimated costs).

i.

Is the heuristics specified in the problem (shown below) admissible? Justify your answer. If the heuristics is admissible, proceed to answer part (ii). If the heuristics is not admissible, correct it with a sensible value of your choice and proceed to answer part (ii). **(10.0 marks)**

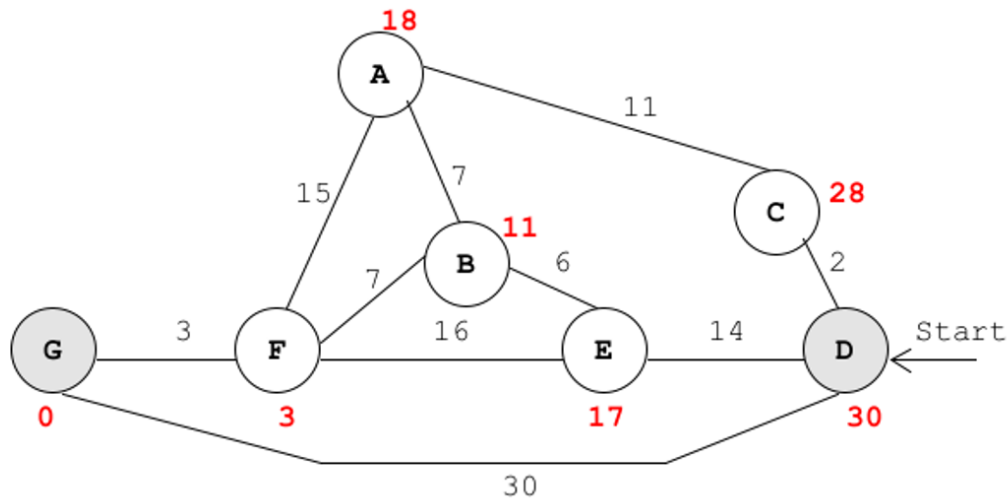


Number in maroon are the heuristics for the respective nodes.
Number in black are the weight (estimated costs).

States	$ h(s) - h(s') $	$C(s, s')$	$h(s) - h(s') \leq C(s, s')$
D-E	$ 30 - 17 = 13$	14	TRUE
C-D	$ 28 - 30 = 2$	2	TRUE
E-F	$ 17 - 3 = 14$	16	TRUE
E-B	$ 17 - 10 = 7$	6	FALSE
B-F	$ 10 - 3 = 7$	7	TRUE
F-G	$ 3 - 0 = 3$	3	TRUE
A-F	$ 18 - 3 = 15$	15	TRUE
A-B	$ 18 - 10 = 8$	7	FALSE
A-C	$ 18 - 28 = 8$	11	TRUE
D-G	$ 30 - 0 = 30$	30	TRUE

The heuristics values at nodes 'E', 'B', and 'A' are not admissible. To correct the heuristics values, we change the heuristics value at node 'B' to 11. With the new revised heuristics value at node 'B' from 10 to 11, the heuristics values are now admissible.

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D-E	$ 30 - 17 = 13$	14	TRUE
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E-F	$ 17 - 3 = 14$	16	TRUE
E-B	$ 17 - 11 = 6$	6	TRUE
B-F	$ 10 - 3 = 7$	7	TRUE
F-G	$ 3 - 0 = 3$	3	TRUE
A-F	$ 18 - 3 = 15$	15	TRUE
A-B	$ 18 - 11 = 7$	7	TRUE
A-C	$ 18 - 28 = 8$	11	TRUE
D-G	$ 30 - 0 = 30$	30	TRUE

The heuristic value of the network is now amissible.

- ii. Perform the A* search to find the shortest path from the start state (A) to the goal state (G). **(35.0 marks)**

The heuristics are admissible now. Hence, we proceed to find the shortest path from node 'D' to node 'G' using the heuristics:

Path expanded	Cost: $f(n) = g(n) + h(n)$	
	$f(D) = g(D) + h(D) = (0) + 30 = 30$	Open: {} Close: {D}
A-B A-C A-G	$f(DG) = g(DG) + h(G) = (0 + 30) + 0 = 30$ $f(DE) = g(DE) + h(E) = (0 + 14) + 17 = 31$ $f(DC) = g(DC) + h(C) = (0 + 2) + 28 = 30$ The winner is the extension to node C. Node 'C' is chosen. The reason node 'C' is chosen because node 'C' is not the goal state, and there is a possibility that the total cost may be lower.	Open: {A,B,C,E,F,G} Close: {D} Open: {A,B,E,F,G} Close: {D,C}
AD-B AD-F	$f(DCA) = g(DCA) + h(A) = (13) + 18 = 31$ $f(DG) = g(DG) + h(G) = (0 + 30) + 0 = 30$ $f(DE) = g(DE) + h(E) = (0 + 14) + 17 = 31$ Since both $f(DCA)$ and $f(DE)$ are higher than $f(DG)$, $f(DG)$ has the lowest cost, and node 'G' is the goal state, hence, the path D-G is the shortest path with the cost of 30.	Open: {B,E,F,G} Close: {D,C,A} Open: {B,E,F,G} Close: {D,C,A,G}

The shortest path found is $D \rightarrow G$ with a total distance of 30.

Notes on Submission

1. This assignment is due on **Saturday, 28 February 2026 by 9:00 pm (21:00 hours) Singapore time.**
2. Submission must be made via **Moodle**. Ensure all files are combined into a single **ZIP file** without subdirectories.
3. Clearly label all components of your submission.
4. Late submissions will incur a **5% deduction per day**, including weekends.
5. Submissions more than **seven** days late will not be marked unless an extension is granted.
6. Extensions must be applied for via **SOLS** before the assignment deadline.
7. Plagiarism is treated seriously. Instances of plagiarism may result in a zero mark for the assignment.

Submit the file **SolutionA3.zip** through Moodle in the following way:

- 1) Access Moodle at **<http://moodle.uowplatform.edu.au/>**
- 2) To login use a Login link located in the right upper corner the Web page or in the middle of the bottom of the Web page
- 3) When successfully logged in, select a site **CSCI203 (SP126) Algorithms and Data Structures**
- 4) Scroll down to a section **Submissions of Assignments**
- 5) Click at **Submit** your Assignment 3 here link.
- 6) Click at a button **Add Submission**
- 7) Move the files created into an area provided in Moodle. You can drag and drop files here to add them. You can also use a link *Add...*
- 8) Click at a button **Save** changes,
- 9) Click at check box to confirm authorship of a submission,
- 10) When you are satisfied, remember to click at a button **Submit assignment.**

A policy regarding late submissions is included in the subject outline. Only one submission per student is accepted.

Assignment 3 is an individual assignment, and it is expected that all its tasks will be solved individually without any cooperation with the other students. Plagiarism is treated seriously. Students involved will likely receive zero. If you have any doubts, questions, etc. please consult your lecturer or tutor during lab classes or over e-mail.

End of specification

If you have any questions or require clarification, please contact your lecturer as early as possible. Good luck!